Fiscal Year:	FY 2008	**Task Last Updated:**	FY 08/06/2008
PI Name:	Dunn-Rankin, Derek Ph.D.		
Project Title:	Applications of Electric Fields in Microgravity Combustion Science (NNX07AB55A)		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Droplets, sprays, particles, dust clouds, nanoparticles		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	ddunnran@uci.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	949-824-8745
Organization Name:	University of California - Irvine		
PI Address 1:	Department of Mechanical & Aerospace Engineering		
PI Address 2:	4200 Engineering Gateway Bldg, EG3224		
PI Web Page:			
City:	Irvine	**State:**	CA
Zip Code:	92697-3975	**Congressional District:**	48
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2001 Physical Sciences 01-OBPR-08
Start Date:	10/01/2006	**End Date:**	09/30/2011
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NASA GRC
Contact Monitor:	Stocker, Dennis P	**Contact Phone:**	216-433-2166
Contact Email:	dennis.p.stocker@nasa.gov		
Flight Program:			
Flight Assignment:	Flight definition. NOTE: Extended to 9/30/2011 per NSSC info (9/2009) NOTE: NEW grant, resulting from grants reinstatement directive in 8/2006, per D. Stocker/GRC (10/07)		
Key Personnel Changes/Previous PI:	A visiting post-doctoral scholar, Dr. Kiyotaka Yamashita has been working on the project this year in the area of numerical modeling; he is participating at no-cost to the project.		
COI Name (Institution):	Weinberg, Felix (Imperial College, London, England) Yuan, Zeng-Guang (National Center for Space Exploration Research, NASA Glenn Research Center)		
Grant/Contract No.:	NNX07AB55A		
Performance Goal No.:			
Performance Goal Text:			

Task Description:

This joint project between the University of California, Irvine and Imperial College, London explores the use of large electric fields in combustion for the purpose of improving performance of energy conversion systems. The project this year focused mainly on preparation for a scheduled spaceflight experiment as well as a requirements review later on this year. Current studies included:
1) Design of a new experimental configuration based on the recommendations of the Science Concept Review (SCR) panel. A substantial portion of development time focused on developing an automated data acquisition scheme so that potential obstacles in automated data acquisition in space could be anticipated.

2) Continued numerical simulations of flame dynamics under the influence of electric fields. The modelling has been expanded to include combustion kinetics.

3) Additional developments in how flame saturation current changes as a function of air dilution into the fuel stream and whether this change can be used to characterize combustion performance.

Task Description:

The overall goal of this project is to understand the behavior of chemi-ionization in flames and to demonstrate the use of this understanding to manipulate combustion in beneficial ways (including enhancing flame stability and reducing flame soot). The project uses detailed measurements of electrical properties of flames in several ways.

First, we can create a neutral wind (ion driven convection) locally at the flame front; this is one of the rare and important action-at-a-distance effects on flames that can be used as a localized actuator. The effect of the wind has been demonstrated with a candle flame in parabolic flight based microgravity and in a spherical cage drop tower, but there has been no systematic study of the ion wind effect in the absence of the confounding effects of the approximately equal influence of buoyancy.

Second, we can use electric fields to affect soot formation and transport – ions and soot are closely linked because there are shared chemical pathways and because soot particles are charged by the flame ions. It is possible, therefore, to modify soot formation and to use charges attached to soot to affect soot transport. By adjusting the soot trajectory through the reaction zone it is possible to affect the oxidation of soot to prevent its eventual release or to modify the residence time of soot in different zones of the flame. It is also possible to use the neutral ion wind to change the mixing behavior of the flame and reduce soot in this way.

Finally, we can use the electrical properties to characterize the flame – there is a quantifiable relationship between the carbon content of a fuel and the ion current it can produce at saturation. This relationship is already exploited in a typical flame ionization detector (FID) analyzer for hydrocarbon emission where the carbon is burned in a hydrogen flame to create an ion current in proportion to the carbon influx. In direct flame interrogation, however, there are several confounding variables that can affect the ion current, including flame temperature (which generally affects the O atom concentration) and the reaction zone volume. Flame temperature can in turn be affected by levels of premixing, diffusion rates, fuel composition, and radiative loss. It is possible, therefore, to determine flame character from the ionization curve response.

Task Significance:

Understanding electric field interactions with flames is central to combustion control, particularly in situations near limit operation of the flame where small effects are amplified dramatically. Electric fields have been shown to modify burning rates as well as to enhance and reduce flame propagation even to the point of flame extinguishment. Our experiments provide a unique ability to make simultaneous measurements of both physical and electrical properties of flames.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

Several of the applications we explore have potential uses in earth gravity. Understanding fundamental electrical properties of flames will also have contributions to earth-based combustion, particularly as regards soot formation and control. Ion current is also a very useful diagnostic tool and sensor for combustion, capable of reflecting flame behavior for next generation power generation systems such as homogeneous charge compression ignition engines. Finally, there are important potential applications in combustion synthesis of nanoparticles where ion formation pathways are important.

Taking the three specific objective topics in turn, the following progress has been achieved over the past year of the project.
Developing a new experimental platform - Along with improving the experimental setup (e.g., we transitioned from using Real Time Windows Workshop, The Mathworks Inc. to the Data Acquisition toolbox package in Matlab), characterization studies proposed for the spaceflight were performed to create a quantitative database of testing conditions. Along with fundamental characterization studies (e.g. VCC curves, step response), we have begun exploring the effect of electric fields on promoting stability of near limit and lifted flames (Science Data End Product D). Calculations on the flow rate required to achieve a lifted flame with the jet diffusion burner showed that an alternative combustion platform was necessary. The CLD Flames group from Yale University (also part of the ACME project) has developed an over-ventilated laminar diffusion flame, which under flow conditions comparable to those planned for the jet diffusion flame are capable of creating a lifted flame. This additional burner configuration allows us to understand the behavior of electric fields over a greater set of flow/flame conditions while taking advantage of the experimental and modeling database available for the coflow burner. We obtained plans for the burner from the CLD Flames group, had the burner constructed, and have begun testing. From the Science Concept Review in March, it was recommended that focus for the spaceflight experiment include this coflow burner. The burner and relevant circuitry for the experiment have been designed, fabricated, and tested.

Baseline testing has revealed interesting and substantial differences between the Voltage-Current Characteristic (VCC) curves of the coflow and the inverted jet flames. With an electric field directed downstream, the behavior of both the jet diffusion and the laminar co-flowing flames initially show similar characteristics. Both burners display a sub-saturation region, where the measured current increases parabolically with electric potential. At a given field strength, which is a

Task Progress:

function of flow conditions, a saturation region is observed where the rate of ion formation is equal to the current collected. At greater field strengths, the behaviors of the two burner types diverge. The measured saturation current of the co-flow flame is constant while the gas jet flame exhibits a region of enhanced ion formation. The measured saturation current for the co-flowing flame without co-flowing air at 100% CH¬4 is 70% the saturation current of the jet diffusion flame at the same flow rate but is only slightly larger than a co-flowing flame with 50% CH4. These results show that the change in geometry causes a reduction in the collected current and removes the super-saturation region. The increased saturation current for the 50% CH4 case with co-flowing air may be the result of the triple flame structure of the co-flowing flame, where a portion of the flame is a lean premixed flame. VCC curves of both burners with various levels of nitrogen dilution can be compared. The jet-diffusion flame has a consistent sub-saturation region regardless of dilution levels; the only difference in the curves is when saturation is achieved. Under the same conditions with a co-flow annulus, the VCC curves of the over-ventilated burner have different slopes but, for the conditions tested, the flames reach saturation at the roughly the same applied potential (~3.6kV), independent of liftoff height. These results present interesting questions regarding the potential effects of geometry. Due to the capillary exit placed flush with the honeycomb annulus, flame instability prevented testing at more than one condition without co-flowing air, making it not possible to isolate the effects of geometry. As a rough confirmation to determine if the capillary flush with the flow straightening honeycomb prevented blowoff, additional tests show that with the capillary raised approximately 25 mm above the flow straightening mesh, the flame could be extinguished by the ion-driven wind. A pure methane flame under increased field strength shows that at -8 kV the flame is oscillating violently, and further increases in field result in flame extinguishment. It is also worth noting that with an increased applied potential; there is a notable reduction in luminous soot

The literature has reported that ion wind effects of a flame when under the influence of applied DC potential are negligible and do not contribute significantly to flame reattachment. However, the conclusions presented in the papers do not include critical measurements of the ion current to understand the contribution, if any, of an ion-driven wind. Experiments on a lifted flame under a DC potential in our laboratory showed a significant reduction in liftoff height. In addition to a study of DC effects on flame liftoff, measurements have also been taken on a lifted flame under an oscillating field.

A significant amount of hysteresis can be observed when a flame is placed under an oscillating field. The maximum ion current observed is also dependent on the forcing frequency. A plot of voltage and the resulting ion current vs. time. The measured current leads the applied voltage, a behavior expected in a capacitor. This behavior, which comes from the capacitance in the electrode mesh not the flame, has been reported in previous work and its presence was compensated for in a closed loop control algorithm created for the jet diffusion burner (F. Borgatelli and D. Dunn-Rankin). Schlieren images of the co-flowing flame with oscillating potentials show that a possible reason for the observed improvement in reattachment is the enhanced mixing caused by the oscillating field creating a fluctuating ion wind.

Numerical Simulation of Small Flames in Electric Fields – Previous numerical simulations have simplified the problem to include static flame shapes with constant properties (e.g. ion number density and flame temperature). These simulations accurately captured experimental measurements of temperature gradients and ion profiles at the grounded mesh for a neutrally buoyant flame. Additional simulations included non-spherical flame shapes to understand the effect of flame shape on the ion flux profile while keeping properties constant (there was little sensitivity to flame shape since the ion field is dominated near the flame by space charge repulsion). Recent developments have been directed toward understanding the mechanism behind the ion current saturation and super-saturation observed in experimental measurements. To further elucidate the complex behavior observed in the flame, a model has been developed by a visiting post-doctoral scholar allowing us to couple combustion kinetics with CFD calculations using PHOENICS/CHEMKIN to simulate the geometry of the jet diffusion flame. Understanding the combustion of methane in the presence of electric fields provides a greater understanding of how ion production is affected by various burner geometries and flow conditions. Focus has been placed initially on the inverted jet diffusion flame due to its broad experimental base with electrical fields, which will allow easy verification of the models performance with our prior measurements.

Simulations on a 2D axisymmetric model of the flame in a downward configuration were run. The flame is created by introducing into an inverted capillary 40 mg/min of methane without dilution. A hot spot of 2000K is placed in the path of the traveling fuel in order to begin the reaction. Using a previously published kinetic mechanism (J. Hu, B. Rivin, and E. Sher, Experimental Thermal and Fluid Science, 21:124–133, 2000), which assumes oxonium (previously called hydronium) ion to be the dominant charged species, simulations were run looking at spatial variations of combustion products under an electric field.

Example results from these calculations include the methane concentration, the ion concentration, the temperature profile of the inverted methane jet for different field conditions, and the flow field for the zero field condition. The temperature surface profiles show that heat is drawn downward as field strength increases as expected from laboratory-based experiments. These results show that although charged species make up a small portion of the total fluid number density, the energy imparted by the electric field, the ion wind effect is an important fluid flow mechanism opposing buoyancy driven natural convection. The most significant results to date from the model are that the simulation can capture the ion current increase with voltage seen in the experiments remarkably well. The absolute value of ion current is in excess by a factor of 2, but the behavior is identical, with the onset of saturation predicted very closely and the super-saturation condition is also seen. We have not yet identified from the model the reason for super-saturation, but it is not due to an absolute increase in ion concentration. Rather, it appears that the ion concentration remains constant while the voltage increases, producing a higher ion flux and higher ion currents. The ability of the flame to continue to source ions at higher field conditions is the reason for super-saturation, and we suspect that this is due to a partial premixing behind the flame sheet by the downward ion wind. The same cannot occur in the co-flow flame because the mesh prevents outside air from breaking into the fuel side of the flame. Naturally, much more work is needed to confirm this mixing hypothesis.

Characterizing Flames by their Saturation Currents – An important conclusion of the earlier phase was that it was certainly possible to distinguish between diffusion and premixed flames by their saturation current - so long as premixed flames were defined in terms of compositions lying between the flammability limits. In fact, one of the mysteries revealed by that work was that, for admixtures of less than 70%, air appears to act as an inert diluent. All the experiments were carried out using a 4mm capillary jet burner, this being the only diameter that could bridge the diffusion and premixed flame regimes. The chief objectives of the current experiments reported here were: 1) the measurement of chemiluminescence; especially CH* emission and 2) relating the total saturation currents, Js, to saturation current densities, js, in the premixed domain and, more generally, to determine fluxes (i.e.

variables/time/area) of other relevant parameters. This requires knowledge of the flame area in addition to the flow rate. In the premixed domain, the flame area, A, can be related to the volume flow rate, Q, directly through the burning velocity, Su.

Three optical filters were used for the chemiluminescence measurements, transmitting at 405 nm, 430 nm, and 515 nm. Initially we attempted to photograph the flame through the CH* (430 nm) filter but there was barely enough luminosity even for the brightest of flames. We might revisit this with a better camera but, as far as we could tell, the geometry of our images did not differ substantially from the corresponding unfiltered luminous shapes. We found that collecting total CH* emission within a constant solid angle is equivalent to measuring total saturation current. In contradistinction to the case of saturation currents, there is no peak in the chemiluminiscence curves corresponding to the premixed regime (just possibly a very small one for C2). However, in interpreting the steady decrease in that region, it should be borne in mind that the flame area rapidly decreases there. Since, at constant volumetric flow rate, the flame area shrinks in inverse proportion to the burning velocity, which reaches a maximum just rich stoichiometric, the CH* emission per unit flame area may, in fact, increase.

For a given reactant composition, temperature and pressure, each premixed flame is characterized by an approximately constant structure orthogonally to its entire area. This is not true of diffusion flames since there is increasing dilution by products in ascending the flame. It could be said that fluxes such as saturation current densities, js, are meaningful only for premixed flames and, conversely, that diffusion flames can be characterized only by the total saturation currents, Js, for a given volumetric flow rate. Therefore, the intermediate region of particular current interest - that rich of stoichiometric composition - presents something of a problem. In the previous report, it was said of this region that, while there is an inner cone visible in photographs at near stoichiometric compositions, there is no conspicuous difference in its appearance from that beyond this limit on the fuel-rich side. This statement had to be reviewed in attempting to measure flame areas. To avoid the tedium of numerical integration, a program that yields surface areas of rotation around a central axis from a contour line “SolidWorks”, a PC based program, was employed. In the case of diffusion flames, the reaction zone was deemed to coincide with the most luminous zone. For premixed flames, the relevant area may depend on the particular parameter investigated.

Burning velocities for premixed methane/air flames are well documented and provide a good starting point. The concept is without significance for diffusion flames, since these do not propagate; it is merely the volumetric flux of reactants. The relevant area for premixed flames is clearly the innermost contour, since the definition of burning velocity, Su, relates to cold unburned reactants. Thus, even the contour of the first onset of luminosity will yield too large an area - i.e. too low an Su value. There is obviously also a margin of uncertainty due to individual judgment as to where to draw the contour. This also applies to the judgment of where the regime changes from diffusion to premixed; i.e. where the transition from maximum luminosity to its first onset is deemed to occur. The other problem lies in the narrowness of the region of the premixed regime within which the burning velocity varies by a factor of about 20. There is virtually a discontinuity in gradient at the peak, so that the maximum value can only be guessed at. Literature values of maximum burning velocities of methane/air mixtures are of the order 35 cm/s (50% higher than our measured values) in the absence of stretch. However, our flames are small (so that average burning velocities will be disproportionately biased by burner rim quenching) and subject to stretch. The peak in ion concentration for an unperturbed flame (one in the absence of applied fields; i. e. in ionization - recombination equilibrium) has been located by probe measurements in premixed flames. It occurs in the main reaction zone. That may not, however, correspond to where the ions are generated. We have used same areas as in the burning velocity measurements, i.e. the centre of the luminous zone for diffusion flames and the first onset of luminosity for premixed flames. This was done partly for convenience but mainly because the resulting maximum saturation current densities still fell below the accurate measurements on flat premixed flames by Lawton and Weinberg. The fractional shortfall was similar to that for burning velocities and analogous arguments apply. The calculation involved tedious interpolations of the original Js curves (see last report), since individual measurements were carried out at different air percentages. Some error may have been incurred in this process, especially around the (somewhat indeterminate) maxima. For diffusion flames, as expected, Js is small and virtually constant at constant fuel flow rate; there may, in fact, be a slight dip as the percentage of air approaches the upper flammability limit. Beyond this point, there is a steep rise in the curve, the values around stoichiometric composition rising by an order of magnitude above those of the diffusion flames.

In determining the CH* per unit flame area, there is again the problematic choice of flame contour defining the area. According to Gaydon, the main reaction giving rise to CH is C2 + OH = CO + CH*. This is consistent with observations from the literature that, for premixed flames, the sequence of the spatial location of the peak intensities tends to be OH*, followed by C2* and then CH* from the unburnt-gas side and this observation is consistent over a range of equivalence ratios. For an ethylene/oxygen diffusion flame, Gaydon finds the same sequence on the fuel side followed by a much larger OH* peak near the maximum temperature. At 1 cm above the base of the Wolfhard - Parker burner, the CH peak is about 1 cm on the fuel side of the maximum temperature. Comparison with ion current measurements suggests that the decrease observed there is largely due to a decrease in flame area. However, the proportional increase in going from diffusion to premixed flames is nowhere near as large as for js and Su. It would be interesting to repeat the calculation for an area close to the contour of maximum luminosity.

Closer examination of the flame photographs suggests that a ring of brighter luminosity occurs within the flame contour. This starts above the burner rim and grows towards the flame apex as the percentage of air increases. Eventually it converges to a point on the axis. After the upper limit of flammability is crossed, this luminous zone detaches from the flame contour and forms an inner cone. If this progression represents the formation of a premixed flame from a pre-reaction zone, which is only stable due to heat and radical transport from the diffusion flame, it may explain why Js does not increase before the upper limit is reached. Pyrolysis due to the adjacent diffusion flame may produce syngas and neither the burning of carbon-monoxide nor that of hydrogen leads to appreciable chemi-ionization. We propose to try shadowgraphy and, if necessary, possibly other optical methods, to establish whether the existence of a premixed pre-reaction zone adjacent to the diffusion flame can be visualized. We plan to pursue parallel numerical simulation of this situation as well. We also propose to measure CH* (and perhaps C2*) emission across the entire range of fuel flow rates and various burner diameters to establish whether the unique dependence of ionization rates on the flux of carbon atoms is paralleled by chemiluminescence at these wavelengths.

Bibliography Type: Description: (Last Updated: 11/26/2025)

Abstracts for Journals and Proceedings	Karnani S, Bennett M, Yamashita K, Dunn-Rankin D. "Recent Progress in Exploring Electric Field Effects on Diffusion Flames." Presented at the The Combustion Institute International Meeting, Quebec, Canada, August 3-8, 2008. Work-in-Progress Poster, The Combustion Institute International Meeting, Quebec, Canada, August 3-8, 2008. , Aug-2008
Papers from Meeting Proceedings	Karnani S, Bennett M, Dunn-Rankin D. "Electrical Properties of Small Diffusion Flames." Western States Section/The Combustion Institute Spring Meeting, University of Southern California, Los Angeles, CA, March 17-18, 2008. Western States Section/The Combustion Institute Spring Meeting, University of Southern California, Los Angeles, CA, March 17-18, 2008. Paper 08S-4. , Mar-2008